AMRITA VISHWA VIDYAPEETHAM

Amrita School of Computing Chennai

Course Delivery Plan

Course Code : Course Name	22AIE205: Introduction to Python	**Program**	B. Tech CSE - AIE
L - T – P - C	1-0-3-2	**Semester / Year**	III/ II
Name(s) of the Faculty	Dr. Prabu M	**Pre-requisite**	Nil
Course Mentor	Dr. Prabu M	**Academic Year**	2026-2027
Course Overview	This course provides a comprehensive introduction to core Python programming, designed to equip students with essential skills for problem-solving and application development. The course emphasizes writing efficient code using Python's control structures, functions, and data structures. It also introduces students to file handling and the modular design of Python applications by building reusable packages and modules. Through hands-on exercises and real-world scenarios, learners will develop the ability to write, test, and package Python programs that are both functional and maintainable. This foundational knowledge serves as a springboard for more advanced programming and software development pursuits.		

Course Objectives		Course Outcomes	
1	To acquire programming skills in core Python.	CO1	Solve problems using Python conditionals and loops.
2	To understand how to write functions and pass arguments in Python.	CO2	Apply Python functions and function calls to solve problems.
3	To develop fundamental understanding of how to build and package Python modules for reusability.	CO3	Apply Python data structures to represent complex data.

4	To develop program in python to read and write files.	CO4	Develop Python Packages for reusability.

Syllabus

Introduction to Python: Control Statements-List, Ranges & Tuples in Python-Python Dictionaries and Sets- Input and Output in Python-Python built in function-Python Object Oriented-Exceptions-Python Regular Expressions-Python Multithreaded Programming-Using Databases in Python-Regular Expression -Thread Essentials-Web Scraping in Python-Data Science Using Python-Graphical User Interface-Django Web Framework in Python Interface of python with an SQL database-Connecting SQL with Python-Performing Insert, Update, Delete Queries using Cursor-NumPy-Pandas and data frame operations on Toyota Corolla Dataset-Data visualization; matplotlib, seaborn libraries-Python Libraries

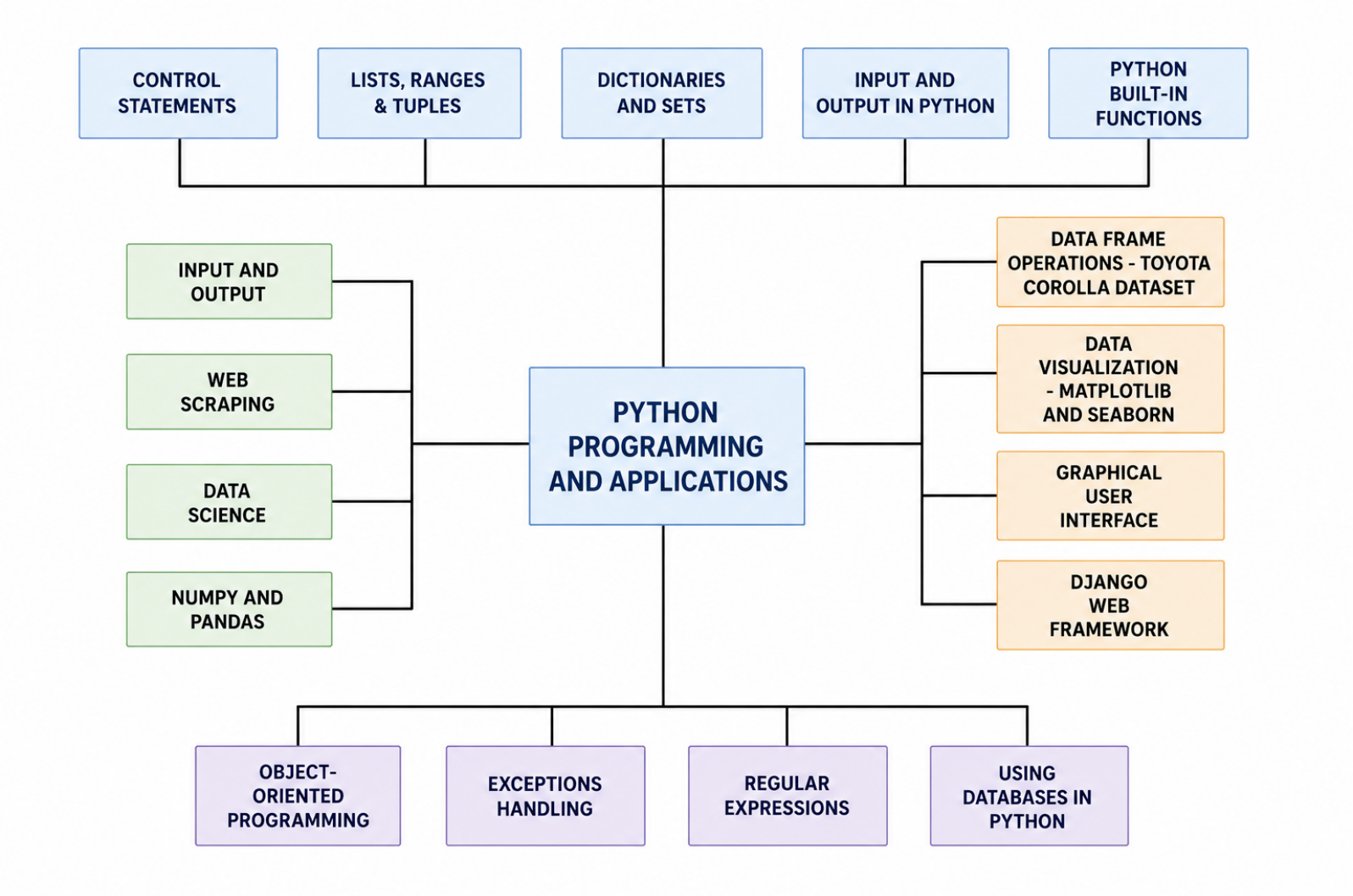
Textbook(s)

T1. Allen B. Downey, “Think Python: How to Think like a Computer Scientist”, 2nd Edition, O’Reilly Publishers, 2016.

Reference(s)

R1. Paul Deitel and Harvey Deitel, “Python for Programmers”, Pearson Education, 1st Edition, 2021.
R2. Eric Matthes, “Python Crash Course, A Hands – on Project Based Introduction to Programming”, 2nd Edition, No Starch Press, 2019.
R3. https://www.python.org/,numpy.org
R4. Martin C. Brown, “Python: The Complete Reference”, 4th Edition, Mc-Graw Hill, 2018.
R5. David Beazley, Brian Jones., “Python Cookbook”, Third Edition, Orelly Publication, 2013, ISBN 978-1449340377

Concept Map


CONTROL STATEMENTS
LISTS, RANGES & TUPLES
DICTIONARIES AND SETS
INPUT AND OUTPUT IN PYTHON
PYTHON BUILT-IN FUNCTIONS
INPUT AND OUTPUT
WEB SCRAPING
DATA SCIENCE
NUMPY AND PANDAS
PYTHON PROGRAMMING AND APPLICATIONS
DATA FRAME OPERATIONS - TOYOTA COROLLA DATASET
DATA VISUALIZATION - MATPLOTLIB AND SEABORN
GRAPHICAL USER INTERFACE
DJANGO WEB FRAMEWORK
OBJECT-ORIENTED PROGRAMMING
EXCEPTIONS HANDLING
REGULAR EXPRESSIONS
USING DATABASES IN PYTHON

Evaluation and Grading					
Internal (70)				**External (30)**	**Total**
Components		**Marks**	**Total Marks**	**End Semester = 30**	**Internal + External = 100**
Mid Term	Mid Term Exam	20	20		
Continuous Assessments	Quiz (2)	5	50		
	Python Foundation Certification – Beginner Level by Infosys Springboard (Assignment-1)	10			
	Assignment-2 Lab Activity Sheets	10			
	Class Participation	05			
	End Sem Lab Assessment	10			
	Mid Sem Lab Assessment	10			

Programme Outcome (PO)	
PO1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identity, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Have the ability to apply mathematical and analytical techniques to model complex problems.
PSO2	Have a strong foundation in programming, together with knowledge of modern languages, tools and technologies needed to build secure, robust software systems.
PSO3	Have the knowledge of AI and ML techniques required for the design and development of intelligent systems to solve real world problems.

CO – PO Affinity Map

PO/PSO CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO 2**	**PSO3**
CO 1	3	3	2	2	3	-	-	-	3	2	3	3	1	1	1

CO 2	3	3	3	3	3	-	-	-	3	2	3	3	1	1	1
CO 3	3	2	3	3	3	-	-	-	3	2	3	3	1	1	1
CO 4	3	2	3	3	3	-	-	-	3	2	1	3	1	1	1

3 – Strong, 2 Moderate, 1 -weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	**Level 2**	**Level 3**
55%	**57%**	**50%**	**60%**	**70%**

Class	**Topics to be covered**	**Mode of Teaching**	**In-Class Activities**	**Out- Class Activities (E – Ref.)**	**CO Mapping**	**Reference**
1 (L)	Introduction to Python	Smart Board and PPT	Student Centric Approach	https://www.youtube.com/watch?v=hEgO047GxaQ&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=2 https://www.youtube.com/watch?v=CScxy0294SE&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=3 https://www.youtube.com/watch?v=PRGkYivK2xI&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=4 https://www.youtube.com/watch?v=BBb_duZIusU&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=5 https://www.youtube.com/watch?v=kU_ZtZhmmEU&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=85	CO1	T1, R1, R2

2-4 (P)	Control Statements	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=NvdSKgQcyuc&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=6 https://www.youtube.com/watch?v=2gFqUWO-AWM&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=7 https://www.programiz.com/python-programming/if-elif-else https://www.programiz.com/python-programming/for-loop https://www.programiz.com/python-programming/while-loop https://www.programiz.com/python-programming/break-continue	CO1	T1, R1, R2
5-7 (P)	List	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=jgNB4GN1UaQ&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=12 https://www.youtube.com/watch?v=9OeznAkyQz4 https://www.programiz.com/python-programming/list	CO3	T1, R2, R4
8-10 (P)	Ranges & Tuples in Python	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=zLFituJxj6c https://www.programiz.com/python-programming/tuple	CO3	T1, R2, R4

				https://www.programiz.com/python-programming/methods/built-in/range https://www.youtube.com/watch?v=YNDgutH6gjs		
11-13 (P)	Python Dictionaries and Sets	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=35dXEQJkFE8&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=13 https://www.youtube.com/watch?v=2IsF7DEtVjg https://www.programiz.com/python-programming/dictionary https://www.programiz.com/python-programming/set https://www.youtube.com/watch?v=t9j8lCUGZXo	CO3	T1, R1, R5
14-16 (P)	Input and Output in Python	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=4OX49nLNPEE https://www.youtube.com/watch?v=kvKvYXXuHoo https://www.programiz.com/python-programming/input-output-import https://www.youtube.com/watch?v=JBc8LLW5KLQ	CO1	T1, R1, R5
17-19 (P)	Python built in function	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=UzOnFDmoJ9w&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=9	CO2	T1, R2, R5

				https://www.programiz.com/python-programming/function https://www.programiz.com/python-programming/methods/built-in		
20-22 (P)	Python Object Oriented	Lab Practice	Interactive Demonstration	https://www.programiz.com/python-programming/object-oriented-programming https://www.youtube.com/watch?v=qiSCMNBIP2g https://www.youtube.com/watch?v=JeznW_7DlB0	CO2	T1, R2, R5
23-25 (P)	Exceptions	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=fhxByMe0mq8&list=PLavw5C92dz9FePZXIo4jpsupqp3X0w2SW&index=11 https://www.youtube.com/watch?v=6SPDvPK38tw&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=70 https://www.programiz.com/python-programming/exceptions https://www.programiz.com/python-programming/exception-handling	CO2	T1, R4, R5
26 (L)	Python Regular Expressions	Smart Board and PPT	POGIL approach	https://www.programiz.com/python-programming/regex https://www.youtube.com/watch?v=V_BozMwoYe4&t=1s	CO3	T1, R4, R5

27 (L)	Python Multithreaded Programming	Smart Board and PPT	POGIL approach	https://www.youtube.com/watch?v=GqHLztqy0PU&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=71 https://www.dataquest.io/blog/multithreading-in-python/ https://www.youtube.com/watch?v=JnFfp81VbOs&t=243s	CO3	T1, R2, R4, R5
28 (L)	Thread Essentials	Smart Board and PPT	Interactive session	https://www.simplilearn.com/tutorials/python-tutorial/python-threading https://www.site24x7.com/learn/threading-in-python.html	CO3	T1, R2, R4, R5
29 (L)	Web Scraping in Python	Smart Board and PPT	Interactive session	https://www.youtube.com/watch?v=XVv6mJpFOb0 https://www.simplilearn.com/tutorials/python-tutorial/web-scraping-with-python	CO4	T1, R2, R4, R5
30 (L)	Data Science Using Python	Smart Board and PPT	Interactive session	https://www.youtube.com/watch?v=hqY0fTUChuc https://www.youtube.com/watch?v=mkv5mxYu0Wk https://www.youtube.com/watch?v=-6RqxhNO2yY https://www.geeksforgeeks.org/data-science-with-python-tutorial/	CO4	T1, R2, R4, R5
31 (L)	Graphical User Interface	Smart Board and PPT	Interactive session	https://www.geeksforgeeks.org/python-gui-tkinter/	CO4	T1, R2, R4, R5

				https://www.simplilearn.com/tutorials/python-tutorial/python-graphical-user-interface-gui https://www.youtube.com/playlist?list=PLCC34OHNcOtoC6GglhF3ncJ5rLwQrLGnV		
32 (L)	Django Web Framework in Python	Smart Board and PPT	Interactive session	https://www.geeksforgeeks.org/python-web-development-django/ https://www.simplilearn.com/tutorials/django-tutorial/what-is-django-python https://blog.jetbrains.com/pycharm/2023/11/what-is-the-django-web-framework/ https://www.youtube.com/watch?v=F5mRW0jo-U4	CO4	T1, R2, R4, R5
33-35 (L)	SQL database-Connecting SQL with Python	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=WDEyt2VHpj4&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=79 https://www.youtube.com/watch?v=vR5utJvN4JY&list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3&index=80 https://www.geeksforgeeks.org/how-to-connect-python-with-sql-database/ https://k21academy.com/datascience-blog/step-by-step-guide-to-setup-sql-with-python/ https://medium.com/learning-sql/python-for-sql-an-introduction-to-database-connectivity-e1a692e16156	CO4	T1, R2, R4, R5

				https://www.youtube.com/watch?v=g60QghtJmjY		
36 (L)	Performing Insert, Update, Delete Queries using Cursor	Smart board and PPT	Direct Instruction Method	https://www.teachoo.com/22106/4582/Performing-insert--update--delete-queries-using-cursor/category/Concepts/ https://www.youtube.com/watch?v=lK-P5kOiQ6Y https://www.geeksforgeeks.org/crud-operation-in-python-using-mysql/	CO4	T1, R2, R4, R5
37-39 (P)	Python Libraries (NumPy & Pandas)	Lab Practice	Interactive Demonstration	https://www.youtube.com/watch?v=FniLzpaSFGk https://www.youtube.com/watch?v=aYmcRnmZVGQ&list=PL9n0l8rSshSnragNblKDBsT8Xu3otp3jA	CO4	T1, R3
40 (L)	Data frame operations on Toyota Corolla dataset	Smart Board and PPT	Direct Instruction Method	https://rpubs.com/Davo2812/1117071	CO4	T1, R3
41 (L)	Data visualization; matplotlib	Smart Board and PPT	Direct Instruction Method	https://www.youtube.com/watch?v=DFBkTIhptOQ&list=PLSwH4ViBDl2R35TpL1IV0vm74xlCc5BEV&index=2	CO4	T1, R3
42 (L)	Seaborn libraries	Smart Board and PPT	POGIL approach	https://www.youtube.com/watch?v=eG663qYKjVw https://www.youtube.com/watch?v=TLdXM0A7SR8	CO4	T1, R3

List of experiments

S. No	Name of the Experiments	CO
1.	Input & Output	CO1
2.	Control Statements	CO1
3.	Lists	CO3
4.	Tuples & Ranges	CO3
5.	Dictionaries, Sets	CO3
6.	Functions	CO2
7.	Python Object-Oriented	CO2
8.	Exception Handling	CO2
9.	Python Libraries (NumPy & Pandas)	CO4

Software Requirements:

1. Anaconda

Attendance Requirements:

(i) All students must maintain a minimum of 75% attendance in each course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.

(ii) Students who fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams. Requests for exceptions will not be considered under any circumstances.

(iii) Attendance will be calculated up to three days prior to the commencement of the mid-term and end-semester exams, as per the academic calendar.

Faculty **Course Mentor** **Chairperson/ Program Chair**